Q585 at the Schur point: there is no deterministic YBE train on Knutson–Tao puzzles to compare with the crystal commutor

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Provenance (PROTOCOL §2.3). **Author:** Clio. **Date:** 12 Nov 2026. **Repository:** `clio-vega/proofs`. **Code commit:** `4641a0145b29`, directory `code-1112-Q585/` (all numbers below were produced by those scripts at that commit). **Registry:** `proofs/registry/q585-ybe-train-vs-commutor.json`. **Not yet sent** (prove session: no email). The send is queued in `OUTGOING-UNSENT.md`.

Abstract

Q585 asked whether the bijection $\mathrm{LR}(\lambda,\mu;\nu) \to \mathrm{LR}(\mu,\lambda;\nu)$ that a Yang–Baxter "train" argument on Knutson–Tao puzzles would induce at the 0/1 (Schur) point equals the Henriques–Kamnitzer crystal commutor. The brief required one check before any comparison: that such a train exists as a *deterministic* map. It does not, at least not as a train built from the puzzle model's own crossing. The crossing of a line from the NW side with a line from the NE side factors through a single edge, so it has rank 3 out of 9 (Proposition 2). It annihilates exactly the input that the equivariant rhombus would carry (Lemma 3). The bundle crossing it generates never has a single output, and it realises the swap $|\lambda\rangle \otimes |\mu\rangle \mapsto |\mu\rangle \otimes |\lambda\rangle$ only on the two constant words (Claim 1, $N \le 5$). So the pre-registered falsifier ("one $c \ge 2$ case with $T \ne C$ and T deterministic") cannot fire. Q585 is *ill-posed at the instrument*. The prediction "yes" is neither confirmed nor refuted.

The comparison side *is* built and checked. The commutor C is computed by two instruments that share no code path: crystal operators, and RSK plus evacuation. They agree on 1308/1308 highest-weight elements. A lesioned C differs on 1298 of them. A separating control (lex-rank matching) differs from C on all 67/67 cells with $c \ge 2$. A second plausible control (reverse-lex matching) agrees with C on all 67, and would have been blind.

1 What was asked, and the gate

Let $A = B(\lambda)$, $B = B(\mu)$ be $\mathfrak{gl}_n$ crystals and ξ the Schützenberger involution. The Henriques–Kamnitzer commutor is

$$\sigma_{A,B}(b \otimes c) = \xi_{B\otimes A}\big(\xi_B(c) \otimes \xi_A(b)\big),$$

and on highest-weight elements it gives a bijection $C : \mathrm{LR}(\lambda,\mu;\nu) \to \mathrm{LR}(\mu,\lambda;\nu)$. (The formula is taken from `0707.2248`, read first-hand at l.855 in BROWSE 2026-11-11.) Q585 asks whether C equals a map T "induced by the YBE train" on puzzles at the 0/1 point. In my reading index, Q585 was posed as *the map a YBE train argument would induce*, and no source in that index defines T. So the brief's §2 gate is the first question: does a deterministic T exist?

A train argument moves a defect through a lattice model by local moves. Each move is an identity of partition functions on a small region with fixed boundary. When every weight is 0 or 1, each identity is an equality of *counts*. It induces a canonical map on configurations only if the

counts are at most 1 on both sides. A count of 2 on each side gives an equality of numbers and no canonical bijection. That is the sense of "deterministic" below.

2 The puzzle model, re-derived and calibrated

Labels are $0, 1, 2$ (with 2 standing for `10`). An up-triangle Δ has (bottom, left/, right\) in $\{(0,0,0),(1,0,2),(2,1,0),(1,1,1),(0,2,1)\}$. A down-triangle ∇ has (top, right/, left\) in the same five triples.

Lemma 1 (calibration)**.** *With ν on the S side read left to right, λ on the NW side read bottom to top, and μ on the NE side read bottom to top, all encoded as* 01*-words in $n \times m$ boxes, the number of fillings equals $c^{\nu}_{\lambda\mu}$ in all* 340 *cases with $n = m = 3$ (all $\lambda, \mu, \nu \subseteq 3 \times 3$ with $|\nu| = |\lambda| + |\mu|$, including $c^{321}_{21,21} = 2$).*

The enumerator (`puzzle.py`) is my own. The LR side counts highest-weight tableau words (`commutor.py`), so the two sides share no code. The orientation convention was *fitted*: of the 16 variants (three side reversals times two readings of ∇) exactly one scores 340/340, and every other scores at most 169/340 (`calib.py`). I record this as a convention fixed by data, not derived. Everything below uses that variant.

The model as lines. Glue each ∇ to the Δ directly above it along their common horizontal edge. This gives a diamond $\Diamond$. Every ∇ is used once, and the only Δ's left over are the N in the bottom row. A diamond is a crossing of two lines: an *SE-line* enters through its upper-left / edge and leaves through its lower-right / edge, and an *SW-line* enters upper-right \ and leaves lower-left \. The SE-lines start on the NW side (carrying λ) and the SW-lines start on the NE side (carrying μ). Each pair crosses at most once, and the bottom-row Δ's fuse one SE-line with one SW-line into a letter of ν. Exchanging λ and μ means exchanging the two bundles, so the only operator in the model that can carry information between them is the diamond crossing.

3 The crossing is not a braiding

Write D_0 for the diamond, as the map $(\mathrm{SE_{in}}, \mathrm{SW_{in}}) \mapsto (\mathrm{SW_{out}}, \mathrm{SE_{out}})$ with $D_0[(\ell\ell, \ell r), (u\ell, ur)] = \#\{x : (x, u\ell, ur) \in \Delta,\ (x, \ell r, \ell\ell) \in \nabla\}$.

Proposition 2. *D_0 is a 0/1 matrix of rank exactly 3 on the 9-dimensional space $\mathbb{Q}^{\{0,1,2\}^2}$. In particular it is not invertible.*

Proof. The upper bound is structural. A filling of the diamond is a choice of the middle horizontal label $x \in \{0,1,2\}$ together with a Δ and a ∇ on it, so $D_0 = \sum_x |\nabla_x\rangle\langle\Delta_x|$ factors through the 3-dimensional label space of that edge, and $\operatorname{rank} D_0 \le 3$. The entries are 0/1 because the Δ triple is determined by (left, right) whenever it exists: no two of the five triples share both side labels. Equality: the inputs $(0,0)$, $(1,0)$ and $(0,2)$ force $x = 0$, $x = 2$ and $x = 1$ respectively. Their output vectors are $\{(0,0),(1,2)\}$, $\{(0,1)\}$ and $\{(1,1),(2,0)\}$, which have disjoint supports, so they are linearly independent. (The full table is printed by `bundle.py`. SymPy gives rank 3, `local.py`.) □

Lemma 3. *D_0 annihilates the input $(SE, SW) = (0,1)$. It also annihilates $(1,2)$, $(2,0)$ and $(2,2)$.*

Proof. A filling needs $(x, u\ell, ur) \in \Delta$. No Δ triple has (left, right) equal to $(0,1)$, $(1,2)$, $(2,0)$ or $(2,2)$. The side pairs that do occur are $(0,0)$, $(0,2)$, $(1,0)$, $(1,1)$ and $(2,1)$. □

Remark 4. The input $(0, 1)$ is where an SE-line carrying 0 meets an SW-line carrying 1. That is exactly the configuration of the Knutson–Tao equivariant rhombus, whose weight $y_j - y_i$ vanishes at the Schur point. This identification is from memory of the KT equivariant rule and was not re-read this session. So the zero row in Lemma 3 is the Schur-point shadow of the equivariant piece. The 0/1 point and an invertible crossing are bought at the same moment, and cannot both be had.

Claim 1 (computed)**.** Let B_N be the bundle crossing in which each of N SE-lines crosses each of N SW-lines once by D_0, acting on pairs of 01-words (s, t) with equal numbers of 1's. For $2 \le N \le 5$:

1. $B_N(s, t)$ is never a single configuration with multiplicity 1. The number of distinct outputs reaches $5, 12, 49, 152$, and B_N kills 1/6, 6/20 and 28/70 of the pairs for $N = 2, 3, 4$.

2. Restricted to 01-outputs, $B_N(s, t)$ equals exactly (t, s) once only for the two constant pairs $s = t = 0^N$ and $s = t = 1^N$, where the swap is the identity.

So the natural train, which carries the λ-bundle across the μ-bundle with the model's own crossing, is not deterministic, and it does not realise the exchange on any input that carries information. This is the content of the gate.

Corollary 5 (Q585 is ill-posed at the instrument)**.** *Within the Schur-point Knutson–Tao model, no train built from its own crossing induces a map* $\mathrm{LR}(\lambda, \mu; \nu) \to \mathrm{LR}(\mu, \lambda; \nu)$. *Hence the comparison "$T = C$?" has no left-hand side, and the pre-registered falsifier cannot fire.*

What this does *not* show. It does not rule out a train using a defect R *foreign* to the model, one that is not a product of puzzle pieces. A sweep over all weight-conserving 0/1 defects satisfying YBE and bootstrap relations with (D_0, Δ) would be the next instrument. I did not run it, because I could not write down which YBE such a defect must satisfy without a source that states the train argument. That source is the reading task Q585 now reduces to (see §5).

4 The commutor side, built and controlled

Instruments. The tensor product $b \otimes c$ is the concatenated word $\mathrm{read}(b)\,\mathrm{read}(c)$, where read reads rows bottom to top. The crystal operators use the bracket rule ($i{+}1 =$ '(', $i =$ ')'). Then $\mathrm{LR}(\lambda, \mu; \nu) = \{b \in SSYT(\lambda) : \mathrm{read}(b)\,\mathrm{read}(u_\mu) \text{ is highest weight of weight } \nu\}$.

C_1: ξ on a connected component by crystal operators only. Walk x up to the highest weight, recording $x = f_{i_1} \cdots f_{i_k} u$. Walk u down to the lowest weight ℓ. Then $\xi(x) = e_{i_1^*} \cdots e_{i_k^*} \ell$, with $i^* = n - i$.

C_2: no crystal operators. ξ on tableaux is evacuation by jeu de taquin. ξ on a word w is the word with $P = \mathrm{evac}(P(w))$ and $Q = Q(w)$, by inverse RSK.

Preliminary checks: C_1's ξ equals evacuation on 405/405 tableaux, and is involutive. Crystal operators commute with RSK (Q preserved, and P moved by the same operator) on 9000/9000 random checks.

Claim 2 (computed)**.** For $n = 4$ and all $1 \le |\lambda|, |\mu| \le 5$ (1241 cells, 1308 highest-weight elements, 67 cells with $c \ge 2$): $C_1 = C_2$ on 1308/1308. Every image has the form $c' \otimes u_\lambda$ with $c' \in \mathrm{LR}(\mu, \lambda; \nu)$ (1308/1308), and $C_{\mu,\lambda} \circ C_{\lambda,\mu} = \mathrm{id}$ (1308/1308).

Controls.

- *Lesion* (drop the outer $\xi_{B\otimes A}$): differs from C_1 on 1298/1308. So the agreement in Claim 2 is not vacuous.
- *A non-control I nearly used:* "apply ξ to each factor, flip, and raise to highest weight" *equals* C identically. $\xi_B(c)\otimes\xi_A(b)$ is the lowest-weight element of its component when $b\otimes c$ is highest weight, and ξ of a lowest-weight element is the highest-weight element. A cheap "flip" is the commutor in disguise.
- *Flip control* F (lex-rank matching of $\mathrm{LR}(\lambda,\mu;\nu)$ with $\mathrm{LR}(\mu,\lambda;\nu)$): differs from C on 67/67 cells with $c \geq 2$, so the sample separates. Reverse-lex matching *agrees* with C on 67/67. With $c = 2$ the two matchings are the two bijections, so this records that C is consistently the anti-lex matching in this range. It is a fact about the sample, not claimed beyond it. Had I chosen reverse-lex as the control, it would have read "$F = C$ everywhere".
- For $\lambda = \mu$, C has no fixed point in any of the 13 cells with $c \geq 2$.

A note on the brief's sample: "$n = 3$ ($\mathrm{Gr}(k,3)$)" carries zero bits, because every Littlewood–Richardson coefficient for $\mathrm{Gr}(k,3)$ is at most 1. The first case with $c = 2$ needs $\nu = (3,2,1)$, so it lives in $\mathrm{Gr}(3,6)$ or in $\mathfrak{gl}_3$ with $|\nu| = 6$. The sample above contains it.

5 What Q585 reduces to

1. A *source* for the train. Q585 asserted "KTW prove $c^\nu_{\lambda\mu} = c^\nu_{\mu\lambda}$ by moving an R-vertex through the puzzle". I could not reconstruct that argument from the model, and the model's own crossing rules it out (above). The KTW work I can name that has a deterministic tropical move is the octahedron-recurrence proof of the LR rule. Henriques–Kamnitzer's octahedron paper (ID recalled as `math/0408114`, *not in my index, unverified*) defines a commutor on hives by the tropical octahedron recurrence. If, as I recall, they prove it equals the crystal commutor, then the hive form of Q585 is a theorem in print, and the remaining question is transport through a puzzle–hive bijection. That is a reading task, graded `recalled`, and nothing in this note rests on it.
2. The C instrument is ready for that comparison: two instruments, a lesion, and a separating control, over 67 cells with $c \geq 2$.

6 Verification summary

item	script	result
puzzle counts $= c^\nu_{\lambda\mu}$	`calib.py`	340/340 (1 of 16 variants; others ≤ 169)
rank D_0	`local.py`	3 (structural bound ≤ 3)
B_N single-output	`bundle.py`	0 pairs, $N = 2..5$
B_N clean swap	`bundle.py`	constant words only
ξ: crystal vs evacuation	`crystal.py`	405/405
RSK–crystal compatibility	`crystal.py`	9000/9000
$C_1 = C_2$	`commutor.py`	1308/1308
lesion differs	`controls.py`	1298/1308
$F_{\mathrm{lex}} \neq C$	`controls.py`	67/67 cells